(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIRST SEMESTER (B.TECH) JANUARY-2024

Paper Code: ES-119 ***Subject: Manufacturing Process***

Time: 3 Hours ***Maximum Marks :60***

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume suitable missing data, if any.

Q1 **Short questions:-** **(2x10=20)**

(a) Explain various constituents of molding sand.
(b) Differentiate between hot working and cold working.
(c) Differentiate among Stiffness, ductility, and toughness with some examples.
(d) How is forging different from casting with respect to properties obtained in a product?
(e) Explain the term DCSP and DCRP in arc welding.
(f) What is spring back in sheet metal.
(g) Why flux is not required in electric resistance welding.
(h) What is edge preparation in Welding.
(i) Differentiate between thermosetting and thermoplastic polymer with some examples.
(j) What is rotational moulding of plastics? Name the parts made by rotational moulding.

UNIT-I

Q2 (a) What is die casting? Explain the difference between hot chamber and cold chamber die casting with neat sketch. **(5)**

(b) Discuss five major casting defects along with their remedies. **(5)**

Q3 (a) Discuss the five types of pattern allowances considered in casting. **(5)**

(b) Explain any five tools used in moulding process with diagrams. **(5)**

UNIT-II

Q4 (a) What is the difference between TIG and MIG welding. Explain with diagrams and their applications. **(6)**

(b) What do you mean by weldability? What are the pros and cons of welded joints over other joints. **(4)**

Q5 (a) Explain with diagram the working, advantages, limitations and applications of submerged arc welding. **(6)**

(b) Explain any four welding defects with its remedies. **(4)**

UNIT-III

Q6 (a) Sketch and explain the difference between hydrostatic extrusion and tube extrusion with their applications. **(5)**

(b) Explain any five tools used in sheet metal shop with diagrams. **(5)**

ES-119

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Q7 (a) With the help of neat sketches explains the difference among the drop, press and machine forging. **(6)**

(b) Explain the difference between embossing and coining giving sketches. **(4)**

UNIT-IV

Q8 (a) Discuss the working principle of compression molding with its pros and cons. What are the various factors affecting compression molding. **(6)**

(b) Differentiate among blending, compacting and Sintering. **(4)**

Q9 (a) Explain conventional single stage plunger type injection molding with its neat sketch. **(5)**

(b) What are the different materials used for processing plastics. Explain any four with its functions. **(5)**

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END TERM EXAMINATION

FIRST SEMESTER [B.TECH.] MARCH 2023

Paper Code: ES-119 **Subject: Manufacturing Process**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions including Q.No.1 which is compulsory. Select one question from each unit.

Q1 a) What is Manufacturing? List out different manufacturing processes. [2]

b) What is the difference between pattern and casting? [2]

c) Explain the effect of polarity on penetration in DC arc welding. [2]

d) Explain the basic principle of resistance welding. [2]

e) What is the difference between open and closed die forging process? [2]

f) What is powder metallurgy? [2]

g) Define the term plastic. What are its characteristics? [2]

h) List out four types of patterns. [1]

UNIT-I

Q2 a) Describe in detail about the importance of manufacturing towards technological and social economic development. [7]

b) Describe properties of materials: Hardness, elasticity and malleability. [3]

c) What are the desirable properties of molding sand? [5]

Q3 a) Why is it important to provide a means of venting gases from the mold cavity? [3]

b) Describe the process of centrifugal casting with neat sketch. What are the advantages? [7]

c) List out the defects in casting process. Explain any three with neat sketch. [5]

UNIT-II

Q4 a) Describe the oxy-acetylene gas welding technique and give the applications. [7]

b) Explain the principle of submerged arc welding with the help of neat sketch. Discuss its applications. [8]

Q5 a) What are differences between TIG and MIG welding processes? [5]

b) List out the defects in welding process. Write causes and remedies of following welding defects: i) Spatter ii) Overlap iii) Undercut [7]

c) What is the difference between filler metal and flux materials? [3]

P.T.O.

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIRST SEMESTER [B.TECH] MARCH 2023

Paper Code: ETME-105 ***Subject: Manufacturing Processes***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Write **short notes** on the followings: - **(2.5x10=25)**

a) What is the difference between core print and chaplet?
b) Give the classification of steel based on carbon percentage.
c) Define ductility and brittleness.
d) What are the characteristics of moulding sand?
e) Function of the gating system.
f) Briefly explain the forging defects.
g) Briefly explain the Extrusion process with its application.
h) Concept of polarity in welding.
i) What is the use of Flux in welding?
j) Differentiate between coining and embossing?

UNIT-I

Q2 Explain the construction and working of the cupola furnace with neat diagram. **(12.5)**

Q3 a) Discuss various heat treatment processes. **(6)**
b) Explain the investment casting process with neat diagram. **(6.5)**

UNIT-II

Q4 Differentiate between hot working and cold working. Explain the types of forging operations **(12.5)**

Q5 Describe with the help of neat diagram explain the marking, measuring, holding and cutting tools used in the fitting shop **(12.5)**

UNIT-III

Q6 a) Explain the principle of electric arc welding with a neat diagram. **(6)**
b) What is the difference between TIG and MIG welding **(6.5)**

Q 7 a) Explain the Submerged arc welding with a neat diagram. **(6)**
b) Differentiate between welding, soldering and brazing. **(6.5)**

UNIT-IV

Q8 What is sheet metal, and what are the different materials used in sheet metal with its application? Explain the sheet metal operations. **(12.5)**

i) Shearing
ii) Drawing
iii) Nibbling
iv) Punching and blanking

Q9 Describe the powder metallurgy process. State the advantage and applications of powder metallurgy. **(12.5)**

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ETME-105
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